

Risk Assessment and Legal Obligations

Risk Assessment

Based on the study material and the guidelines provided by the EU AI Act, I concluded that our project idea of the AI system developed for predicting traffic accident severity and identifying high-risk intersections in Breda falls into the 'High-risk' category due to its potential impact on safety and livelihood.

High-Risk AI System Classification Explanation

Social Impact and Public Safety:

- The AI system influences road safety measures, impacting law enforcement and emergency service actions, which are crucial for public safety.
- The system's predictive capabilities can significantly affect how and where resources are allocated, potentially impacting the safety and well-being of residents and visitors in Breda.
- Since our project aims to predict accident severity preemptively, which could significantly influence driver behavior.

Essential Services and Infrastructure:

- The AI system manages essential public infrastructure by predicting accident hotspots and severities, facilitating targeted interventions and efficient mobilization of emergency services.

Law Enforcement:

- The system assists in law enforcement by providing data-driven insights for proactive accident prevention and efficient response strategies, affecting public safety directly.

In general, this AI system directly impacts road safety, which is critical for human wellbeing. Providing real-time alerts to users can shape their behavior and decisions, potentially leading to serious consequences if predictions are inaccurate. Inaccurate predictions could result in unintended outcomes, increasing rather than reducing accident risks.

To accurately assess the risks associated with our AI system, understanding its impact on public safety and accident-involved individuals is crucial. Continuous monitoring and evaluation of our AI model's performance and impact allow us to identify risks and vulnerabilities. This iterative process enables us to adapt and improve our model, minimizing potential risks.

Legal Obligations

The AI system is categorized as high-risk due to its predictions potentially impacting safety measures, emergency services distribution and law enforcement actions. Thorough explanation and reasoning is provided in the section above.

Legal Obligations Under the EU AI Act

The SmartAccidentPredict project must adhere to several legal obligations specified in the EU AI Act. Below are the relevant titles and articles, along with our approach to addressing these legal obligations during the implementation process:

Title II: Prohibited AI Practices (Article 5)

Obligations:

- Ensure the AI system does not engage in manipulative behavior that exploits vulnerabilities of specific groups (e.g., children).
- Avoid any form of social scoring that categorizes individuals based on personal traits.
- Do not implement biometric identification or mass surveillance practices without proper consent.

Approach:

- The AI system will strictly focus on predicting traffic accident severity and identifying high-risk intersections, ensuring it does not manipulate behavior or exploit vulnerabilities.
- The system will not engage in social scoring or track individuals, maintaining ethical standards and respecting user privacy.
- There will be no use of biometric identification or mass surveillance techniques in the system.

Title III: High-Risk AI Systems (Articles 6-15; Article 16)**Obligations:**

- Classify the AI system correctly as high-risk and ensure it meets all specific requirements outlined for high-risk AI systems.
- Implement risk management systems, ensuring thorough assessment and mitigation of risks associated with the AI system.
- Maintain data governance, ensuring high-quality data handling, integrity, and accuracy.
- Ensure human oversight and establish processes to allow human intervention in case of system malfunctions or unintended behaviors.
- Conduct regular testing and validation to ensure the AI system performs as intended and remains compliant with safety and accuracy standards.

Approach:

- The AI system will be classified as high-risk due to its impact on public safety and essential infrastructure management.
- A comprehensive risk management framework will be developed, including risk assessments and mitigation strategies throughout the project lifecycle.
- Data governance protocols will be implemented to ensure data quality, security, and integrity, with regular audits and checks.
- Human oversight mechanisms will be established, including protocols for manual intervention and decision-making.
- Regular testing and validation of the AI system will be conducted, with performance metrics tracked and reported.

Title IV: Transparency Obligations for Providers (Article 52)**Obligations:**

- Inform users that they are interacting with an AI system and provide clear information about its capabilities and limitations.
- Ensure transparency in AI-generated content, making it clear when content is created or influenced by AI.

Approach:

- The interactive dashboard will include clear notices that predictions and analyses are AI-generated.
- Documentation and user guides will be provided to explain the AI system's capabilities, limitations, and how to interpret its predictions.
- Any content or insights generated by the AI will be transparently labeled as AI-generated.

Title VIII: Post-Market Monitoring (Article 61; Article 62)**Obligations:**

- Implement post-market monitoring procedures to ensure ongoing compliance, safety, and reliability of the AI system.
- Regularly update the system to maintain accuracy and prevent bias.
- Establish processes for reporting and addressing issues or malfunctions that arise post-deployment.

Approach:

- A post-market monitoring plan will be developed, including continuous performance tracking, compliance audits, and user feedback mechanisms.
- The AI system will undergo regular updates and recalibration to maintain accuracy and address any biases.
- A reporting mechanism will be established for users to report issues, with a dedicated team in place to address and resolve any problems.

By adhering to these legal obligations, the SmartAccidentPredict project aims to ensure that the AI system is developed and deployed ethically, transparently, and safely. Our approach includes comprehensive risk management, data governance, transparency, human oversight, and continuous monitoring to meet the requirements of the EU AI Act and maintain public trust in the system.

To fulfill these responsibilities, comprehensive documentation of the AI system's development, data origins, and decision-making methods must be kept. Users should undergo training to grasp the AI system's capabilities and limitations. Furthermore, ongoing monitoring is essential to continuously update and evaluate the system for accuracy, bias, and potential harm.

My reflection

For the SmartAccidentPredict project, my contributions to the legal assessment included thorough research, documentation, and implementation planning in compliance with the EU AI Act. My primary goal was to ensure that the project adhered to essential legal frameworks, with a specific emphasis on transparency and classification of high-risk factors. This approach guarantees that the project meets both legal mandates and ethical guidelines, promoting the responsible deployment of AI technology used within our dashboard. Below are the evidences of my contributions and reflections on the legality of individual contributions made to the project:

Research and Documentation

- I reviewed the relevant titles and articles of the EU AI Act, specifically focusing on Titles II, III, IV, and VIII.
- I summarized the legal obligations for our team, highlighting the necessary compliance measures for each relevant article.
- I documented the steps required to ensure our AI system adheres to the prohibited practices, high-risk classifications and requirements, transparency obligations, and post-market monitoring.

Reflection: Understanding the AI Act's requirements was crucial for guiding the project's direction and ensuring compliance. This research provided a foundation for our team to integrate legal considerations into our technical and operational plans from the outset. Each of the members contributed to the legal assessment in a similar way individually.

Reflection on the Legality of Individual Contributions

Each team member's contributions were assessed to ensure they aligned with the legal requirements of the EU AI Act. Below are reflections on the legality of these contributions:

Data Collection and Pre-processing (Kees, Michal, and Viktoria)

- Legal Assessment: Ensured data privacy and integrity during the collection and pre-processing stages. Complied with data governance standards to maintain high data quality.
- Reflection: Proper data handling is critical for the accuracy and legality of the AI model. This stage laid the groundwork for compliant data use throughout the project.

Model Development and Training (Kees, Michal, and Viktoria)

- Legal Assessment: Implemented risk management strategies during model development to mitigate potential biases and ensure ethical use of AI.
- Reflection: Developing the model with legal and ethical considerations in mind ensured that the AI system was both effective and compliant with high-risk AI system requirements.

Interactive Dashboard Development (Jedrzej)

- Legal Assessment: Ensured the dashboard was transparent about its AI functionalities and provided users with clear information on AI-generated content.
- Reflection: Transparency in the user interface is crucial for user trust and legal compliance, particularly under the transparency obligations of the AI Act.

Summary

My contributions to the legal assessment focused on ensuring comprehensive compliance with the EU AI Act through detailed research, planning, and implementation. Each team member's work was assessed for legality, ensuring that our project not only met technical goals but also adhered to legal and ethical standards. This holistic approach was critical in developing a responsible and compliant AI system for the SmartAccidentPredict project.